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Confident Misalignment, Horizon-Framed: A Structural Account of Agentic Failure

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Abstract

The dominant failure mode of agentic systems was named by the HGC³AE² framework paper (MR-P1) as **confident misalignment**: outputs that are coherent, plausible, and authoritative while nevertheless diverging from validated reality or from the user's actual intent. The MR-P1 framing is canonical and is not redefined by this paper. What this paper supplies is a **structural account** of the failure: confident misalignment, read through the intent horizon developed in IH-1 and the intent-CIA-NR model developed in IH-2, is the operational state in which an agentic system **operates outside its intent horizon while reporting horizon-integral status to its operator**. This re-description makes the failure mechanically diagnosable rather than retrospectively recognizable.

The argument is in four movements. Section 2 establishes the cite-without-redefinition scope: MR-P1's definition of confident misalignment is canonical; this paper supplies an additional structural account that operates alongside the canonical definition, not in place of it. Section 3 develops the horizon-framed account: confident misalignment, in horizon vocabulary, is the joint condition of intent-integrity loss (the system has drifted outside the operator-pinned envelope) and intent-availability illusion (the system's self-reports still indicate the prior envelope). Section 4 specifies the three structural varieties of horizon-framed confident misalignment — substrate-boundary breach, conceptual-boundary drift, governance-boundary stale-pin — and the diagnostic signature of each. Section 5 names what intent-CIA-NR loss looks like at each boundary, anticipating IH-4's diagnostic instruments and IH-5's enforcement gates. Section 6 closes with implications for operator practice and downstream-series inheritance.

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ABSTRACT

The dominant failure mode of agentic systems was named by the HGC³AE² framework paper (MR-P1) as **confident misalignment**: outputs that are coherent, plausible, and authoritative while nevertheless diverging from validated reality or from the user's actual intent.[1] The MR-P1 framing is canonical and is not redefined by this paper. What this paper supplies is a **structural account** of the failure: confident misalignment, read through the intent horizon developed in IH-1[2] and the intent-CIA-NR model developed in IH-2,[3] is the operational state in which an agentic system **operates outside its intent horizon while reporting horizon-integral status to its operator**. This re-description makes the failure mechanically diagnosable rather than retrospectively recognizable.

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The central claim is that **the dominant failure mode of agentic systems is not a model property; it is a horizon-governance gap**. A system in a state of confident misalignment is not malfunctioning at the model level — it is operating competently against an envelope that no longer matches the operator's pin. The structural account makes the failure mechanically detectable; the model-level account (which MR-P1 supplies) makes the failure conceptually nameable. Both accounts are required; this paper supplies the structural half.

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1 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285. **Cited as canonical for the definition of confident misalignment; the present paper does not redefine the term, only supplies a structural account that operates alongside the canonical definition.**

2 Justin H. Kuiper, CISSP, *The Intent Horizon: A Structural Primitive for Governed Agentic Operation* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-horizon/.

3 Justin H. Kuiper, CISSP, *Intent Integrity: CIA and Non-Repudiation as Properties of Intent, Not Data* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-integrity/.

1. INTRODUCTION

1.1 SCOPE DISCLAIMER (CITE-WITHOUT-REDEFINITION)

This paper cites the HGC³AE² Framework paper (MR-P1)[4] as the canonical definition of confident misalignment. **This paper does not redefine confident misalignment.** Per the H-21 cite-without-redefinition discipline applied in the *Intent Horizon Theory* series, the original definition stands as canon; the contribution of this paper is a *structural account* — a horizon-framed re-description that operates alongside the canonical model-level definition without replacing it. Readers familiar with MR-P1 should expect the present paper to extend, not amend, that work.

The cite-without-redefinition discipline matters in this paper specifically because confident misalignment is the load-bearing diagnostic concept of the entire YKS canon. Any apparent redefinition of the term creates ambiguity downstream — particularly for the Cyber Governance (#348) and Governed Autonomy (#349) series whose entire frame depends on the term's settled meaning. This paper is structured to leave MR-P1's definition undisturbed while making the term's diagnostic application more precise.

1.2 WHAT THIS PAPER SUPPLIES

The MR-P1 framing established confident misalignment as a *model-level* phenomenon: outputs that are coherent, plausible, and authoritative while diverging from validated reality or user intent. The framing names the failure conceptually and makes the case for HGC³AE² as the architectural response. What the framing does not supply is a structural account of *what state the system is in* when it is producing confidently misaligned output. The IH-1 and IH-2 papers supply the vocabulary in which the structural account becomes specifiable.

Structurally, the system in confident-misalignment state is:

Outside its intent horizon along at least one of the four boundary dimensions (temporal, conceptual, substrate, governance);[5]

Operating with intent-integrity loss (the operating envelope no longer matches the operator's pinned commit);[6]

Reporting intent-availability that indicates the operator-pinned envelope (the system's introspective query returns the pinned state, not the drifted operating state).

The combination — drift in fact, no drift in self-report — is the structural signature of confident misalignment. The signature is *mechanical*: it can be detected by comparing operator-side queries (intent-availability) against chain-of-custody attestation (intent-integrity) and finding the mismatch.

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4 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285. **Cited as canonical for the definition of confident misalignment; the present paper does not redefine the term, only supplies a structural account that operates alongside the canonical definition.**

5 Justin H. Kuiper, CISSP, *The Intent Horizon: A Structural Primitive for Governed Agentic Operation* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-horizon/.

6 Justin H. Kuiper, CISSP, *Intent Integrity: CIA and Non-Repudiation as Properties of Intent, Not Data* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-integrity/.

1.3 WHAT THIS PAPER DOES NOT CLAIM

This paper is not arguing that the MR-P1 framing is incomplete. The model-level framing remains canonical for the question *what is confident misalignment*. This paper addresses an adjacent question: *what state is the system in when it is in that failure*? Both questions have answers; the answers are different. This paper supplies the structural answer; MR-P1 supplied the conceptual one.

Section 2 establishes the relationship between the model-level and structural accounts in detail. Section 3 develops the horizon-framed account: confident misalignment as the joint intent-integrity-loss + intent-availability-illusion state. Section 4 specifies the three structural varieties and their diagnostic signatures. Section 5 maps each variety to the intent-CIA-NR properties and anticipates the enforcement gates IH-5 will name. Section 6 hands off to IH-4 (Epistemic Weather) where the diagnostic instruments operationalize the structural account.

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2. TWO ACCOUNTS OF ONE FAILURE

2.1 THE MODEL-LEVEL ACCOUNT (MR-P1, CANONICAL)

MR-P1 names confident misalignment as a property of agentic-system outputs: coherent, plausible, authoritative outputs that diverge from validated reality or from operator intent.[7] The account is model-centric: the failure is attributed to the agentic model's reasoning behavior, the structural responses are governance, context, alignment, runtime verification. The account is correct. It is also incomplete for diagnostic purposes — it tells the operator what the failure *is* without telling the operator what the system's *current state* is when the failure is occurring.

2.2 THE STRUCTURAL ACCOUNT (THIS PAPER)

The structural account names confident misalignment as a property of the system's *operating state* relative to its intent horizon: the system is outside its pinned envelope while its self-reports indicate it is inside. The account is horizon-centric: the failure is attributed to the divergence between operating envelope (where the system actually is) and reported envelope (where the system says it is). The structural responses are intent-CIA-NR enforcement (§5) and horizon-state instrumentation (IH-4).

2.3 HOW THE TWO ACCOUNTS COMPOSE

The two accounts compose in the following way:

MR-P1 explains *what makes confident misalignment important*: it is the failure mode whose architectural response is HGC³AE², the failure that distinguishes governed agentic operation from capable-but-ungoverned operation.

The structural account explains *what state the system is in during the failure*: outside intent horizon along ≥1 boundary, intent-integrity loss without intent-availability drop.

A practitioner explaining the failure to a colleague uses both accounts. A diagnostic instrument detecting the failure uses the structural account. A governance ceremony deciding

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7 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285. **Cited as canonical for the definition of confident misalignment; the present paper does not redefine the term, only supplies a structural account that operates alongside the canonical definition.**

whether to re-pin uses the model-level account to frame the discussion and the structural account to identify which boundary was breached.

The two accounts are not in tension. They are operating at different levels of abstraction against the same phenomenon, and the architecture is more diagnostically precise when both are available than when only one is.

2.4 WHY THE STRUCTURAL ACCOUNT IS LOAD-BEARING FOR THE REST OF IH-THEORY

IH-4 (Epistemic Weather) specifies operator-facing diagnostic instruments. The instruments must be calibrated against *something*: a set of observable states that the operator reads. Without the structural account, the instruments calibrate against a model-level concept that is hard to operationalize. With the structural account, the instruments calibrate against horizon-boundary state — observable, comparable, time-bounded. IH-5 (Skipjack as Horizon Governance) specifies enforcement gates. The gates fire on *something*: a signal that crosses a threshold. The structural account names what crossings happen, in horizon-boundary vocabulary, that the gates correspond to. Without the structural account, the gates fire on heuristics; with the structural account, they fire on architectural signals.

3. THE HORIZON-FRAMED ACCOUNT

3.1 THE JOINT CONDITION

Confident misalignment, in horizon vocabulary, is the **joint condition** of two states:

Intent-integrity loss. The system's operating envelope has drifted outside the operator's pinned horizon along at least one of the four boundary dimensions. Section 4 develops the three operationally distinct varieties of this drift.

Intent-availability illusion. The system's self-reports about its operating envelope — produced by its introspection mechanism (the intent-rehearsal function[8]), surfaced via intent-availability queries — return the pinned envelope, not the drifted operating envelope.

The conjunction is the failure. Either condition alone is recoverable:

Drift with accurate self-report is the *signal* the architecture is designed to handle: the system reports drift, the operator re-pins or accepts the drift, governance proceeds.

No drift with stale self-report is benign noise: the report does not match recent observations but the operating envelope still matches the pin, so the discrepancy resolves on next query.

The conjunction — drift AND inaccurate self-report — is the failure that confident misalignment names. The system is *confident* (it self-reports horizon-integral status) and *misaligned* (it is operating against a different envelope). The two halves of the term map cleanly onto the two halves of the structural condition.

3.2 WHY THE JOINT CONDITION IS HARD TO DETECT

Several reasons:

8 Justin H. Kuiper, CISSP, *The Comprehension Instrument for Agentic Systems* (v0.1-seed, 2026), nonsequitur.tech/white-papers/comprehension-instrument/.

The system's introspection mechanism is producing output that is internally consistent — by every metric internal to the model, the system is performing well.

The system's outputs are coherent against the operating envelope (the drifted one), so the outputs themselves look like governed operation.

The operator's instruments, if calibrated only on outputs, see governed operation; if calibrated only on introspection, see horizon-integral self-report. Neither instrument independently catches the conjunction.

What catches the conjunction is **chain-of-custody attestation cross-checked against intent-availability query** — the intent-CIA-NR verification primitives developed in IH-2.[9] When the chain-of-custody trace from the execution layer reports an operating envelope different from the intent-availability query result, the conjunction is structurally visible. The discrepancy is mechanical, not interpretive: it does not require judgment to detect; it requires the verification primitives to be wired together.

3.3 THE TEMPORAL PATTERN

Confident misalignment is rarely a sudden state. The structural account names the typical temporal pattern:

t=0: operator pins horizon at governance ceremony. System operating envelope matches pin.

t=1..N (drift accumulation): micro-decisions in agentic execution accumulate. Operating envelope drifts incrementally; each individual decision is locally defensible.

t=N (boundary crossing): cumulative drift crosses one of the four horizon boundaries. From this point forward, the system is operating outside its horizon.

t=N..M (reporting lag): the system's introspection mechanism has not yet caught up to the drift. Self-reports continue to indicate the pinned envelope.

t=M (detection or escalation): either the structural verification fires (chain-of-custody trace cross-checked against intent-availability returns mismatch) or an external symptom surfaces (an output produces a result clearly outside the pinned envelope and the operator catches it manually).

The architectural goal is to make t=M as close to t=N as possible. Without the structural account, t=M depends on operator vigilance and is highly variable. With the structural account + IH-4's instruments + IH-5's enforcement gates, t=M is mechanical and closer to t=N.

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4. THREE STRUCTURAL VARIETIES

The horizon has four boundary dimensions. Confident misalignment can occur via any of them, but three structural varieties cover the operationally common cases. Each variety has a diagnostic signature distinguishing it from the others; the IH-4 epistemic-weather instruments use the signatures to classify the active failure.

9 Justin H. Kuiper, CISSP, *Intent Integrity: CIA and Non-Repudiation as Properties of Intent, Not Data* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-integrity/.

4.1 SUBSTRATE-BOUNDARY BREACH

Structural state. The system is operating against a substrate that no longer matches the substrate the pinned horizon was committed against. Common causes: model version shift, context-window expansion or compression, tool-set change, persistence-tier failover, observability-plane disruption.

Failure signature. The system continues to produce outputs as if the substrate were unchanged. Intent-rehearsal returns the pinned envelope. Operations against the new substrate produce outputs whose calibration is degraded *without the calibration drop being self-reported*. The substrate-boundary breach is invisible at the model level and visible only when chain-of-custody attestation is cross-checked against intent-availability query: the chain trace shows the new substrate; the query indicates the old envelope.

Recovery. Re-pin the horizon to the new substrate, or escalate to operator for governance-ceremony review. The architectural response is the same as for any substrate event; the contribution of the structural account is that the response now has a *trigger* (the boundary breach) and a *signal* (the cross-check mismatch).

4.2 CONCEPTUAL-BOUNDARY DRIFT

Structural state. The system is operating against a problem class outside the conceptual scope the pinned horizon committed. Common causes: ambiguous task instructions interpreted at margins, scope creep through follow-up questions, drift through multi-step decomposition.

Failure signature. Each individual step is locally within scope; the cumulative scope after N steps is outside the pinned envelope. The system's self-reports indicate continued operation within scope (because each individual step was within scope at the time it was committed). Chain-of-custody attestation across the N steps shows the cumulative drift; the intent-availability query does not surface the cumulative state.

Recovery. Structural-decomposition fail-closed (Skipjack mechanism, anticipated by IH-5[10]) catches this at the boundary if the cumulative-drift threshold is configured. Without the threshold, conceptual-boundary drift produces the worst-case confident misalignment: long-horizon competent execution against an envelope no one re-pinned.

4.3 GOVERNANCE-BOUNDARY STALE-PIN

Structural state. The system is operating against an intent pinned by a prior governance authority that is no longer the current authority. Common causes: operator shift change, delegation event, persona handoff across accounts.[11]

Failure signature. The system's self-reports return a pinned envelope. The pinned envelope is authentic — it really was pinned by an operator. The problem is that the pinning operator is no longer the relevant governance authority for the current execution context. The non-repudiation log (IH-2 property)[12] reveals the gap: the signature on the active pin does not match the signature of the current governance authority.

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10 Forthcoming IH-5 *Skipjack as Horizon Governance* (in concurrent drafting; this dispatch).

11 Justin H. Kuiper, CISSP, *Cross-Account Harness Coordination: Multi-Account Persona Ops* (v0.1-seed, 2026), nonsequitur.tech/white-papers/cross-account-harness/.

12 Justin H. Kuiper, CISSP, *Intent Integrity: CIA and Non-Repudiation as Properties of Intent, Not Data* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-integrity/.

Recovery. Re-pin under current authority. The architectural response is identical to a new task: governance ceremony, operator review, fresh pin. The structural account makes the recovery *trigger* visible — the gap between the active pin's signature and the current authority's signature is mechanically detectable.

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5. INTENT-CIA-NR MAPPING

Each variety of horizon-framed confident misalignment maps to a specific intent-CIA-NR property loss developed in IH-2:[13]

Variety

Primary intent-CIA-NR loss

Secondary loss

Substrate-boundary breach

Intent-integrity (operating-envelope drift)

Intent-availability (self-report stale)

Conceptual-boundary drift

Intent-integrity (chain-of-custody trace shows drift)

Intent-availability (query indicates pinned)

Governance-boundary stale-pin

Intent-non-repudiation (signature mismatch)

Intent-availability (operator cannot recover current authority's intent)

The mapping is mechanical because the intent-CIA-NR properties are mechanically verifiable. The IH-4 paper specifies the diagnostic instruments by which operators read each property's state in real time;[14] the IH-5 paper specifies the enforcement gates that fire on each property's threshold crossing.[15]

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13 Justin H. Kuiper, CISSP, *Intent Integrity: CIA and Non-Repudiation as Properties of Intent, Not Data* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-integrity/.

14 Forthcoming IH-4 *Epistemic Weather* (in concurrent drafting; this dispatch).

15 Forthcoming IH-5 *Skipjack as Horizon Governance* (in concurrent drafting; this dispatch).

A note on the absent fourth variety: temporal-boundary breach. A pure temporal-boundary breach — the system continuing operation past the temporal duration the operator committed — is *self-detecting* if the system's clock is trustworthy and the governance ceremony pinned an explicit temporal boundary. The architecture's response is escalation, not silent recovery. Temporal-boundary breach produces confident misalignment only when paired with another boundary breach (substrate, conceptual, governance); the present taxonomy collapses such cases into the dominant breach.

6. IMPLICATIONS AND CONCLUSION

The structural account of confident misalignment has implications for operator practice, instrument design, and downstream-series inheritance.

Operator practice. Operators who learn to read structural signals — chain-of-custody attestation mismatch, intent-availability query staleness, non-repudiation signature gap — can detect confident misalignment events earlier than operators relying on output-level review. The instruments do not replace operator judgment; they extend operator reach across a temporal window the prior canon could not specify.

Instrument design. Diagnostic dashboards for governed agentic systems should surface the three structural varieties as named states. The current generation of agentic-system observability tooling surfaces task progress, model performance, and audit logs; the structural account names *which observable states the dashboards should add* for confident-misalignment readability.

Downstream-series inheritance. The #348 (Cyber Governance) and #349 (Governed Autonomy) series cite confident misalignment as the failure mode their entire frame protects against. Until this paper, those series cited MR-P1's model-level account, which is correct but operationally underspecified. With the structural account available, the downstream series can cite the conjunction (intent-integrity loss + intent-availability illusion) directly and operationalize their threat models accordingly.

The central claim is that **confident misalignment is mechanically diagnosable when read structurally** — and mechanical diagnosability is the precondition for operationally responding to the failure before it produces irreversible consequences.

This paper closes Paper Three of the *Intent Horizon Theory* series. IH-4 (Epistemic Weather) operationalizes the structural account through diagnostic instruments — four weather classes the operator continuously reads.[16] IH-5 (Skipjack as Horizon Governance) names the enforcement gates by which the architecture fires on structural-signal threshold crossings.[17]

REFERENCES

— end v0.1-seed draft —

16 Forthcoming IH-4 *Epistemic Weather* (in concurrent drafting; this dispatch).

17 Forthcoming IH-5 *Skipjack as Horizon Governance* (in concurrent drafting; this dispatch).

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